

Association Rule Learning: Market Basket Analysis of the Groceries Dataset

KAGAME Fred

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Question 1: Understand the Data

The Groceries dataset contains **9,835 transactions** across **169 unique items**. Table 1 and Figure 1 show the 10 most frequently purchased items.

Table 1: Top 10 most frequently purchased items.

Item	Count	Support (%)
Whole milk	2513	25.55
Other vegetables	1903	19.35
Rolls/buns	1809	18.39
Soda	1715	17.44
Yogurt	1372	13.95
Bottled water	1087	11.05
Root vegetables	1072	10.90
Tropical fruit	1032	10.49
Shopping bags	969	9.85
Sausage	924	9.40

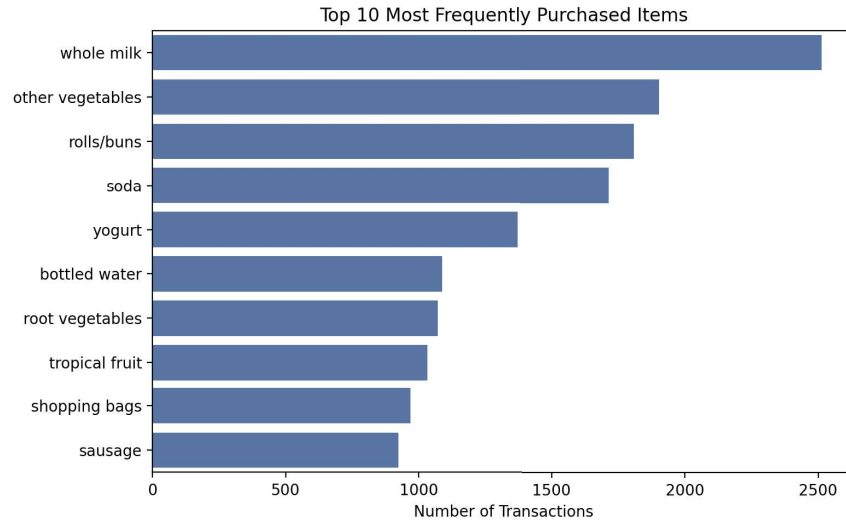


Figure 1: Top 10 most frequently purchased items.

Question 2: Frequent Itemsets

Using the Apriori algorithm with a minimum support threshold of 0.005, at least 10 frequent itemsets of size two or more were identified. Table 2 and Figure 2 show the top 10, ranked by support.

Table 2: Top 10 frequent itemsets containing 2+ products.

Itemset	Support
{whole milk, other vegetables}	0.0748
{whole milk, rolls/buns}	0.0566
{whole milk, yogurt}	0.0560
{whole milk, root vegetables}	0.0489
{root vegetables, other vegetables}	0.0474
{yogurt, other vegetables}	0.0434
{other vegetables, rolls/buns}	0.0426
{whole milk, tropical fruit}	0.0423
{whole milk, soda}	0.0401
{soda, rolls/buns}	0.0383

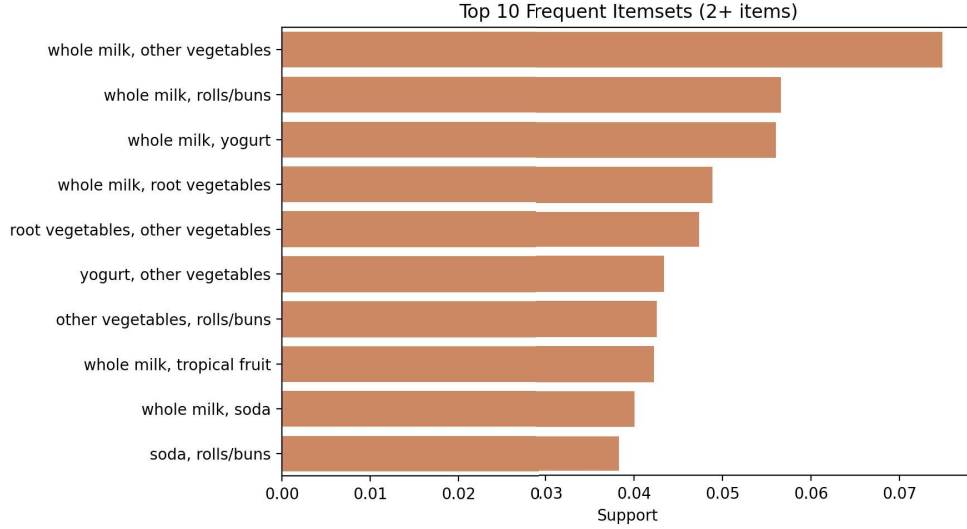


Figure 2: Top 10 frequent itemsets by support.

a–c. The itemset with the highest support is **{whole milk, other vegetables}**, with support ≈ 0.0748 — meaning these two items are bought together in about 7.5% of all transactions, the most common two-item combination in the store.

d. What the support value tells us: support reflects how common a combination of items is across all customers, independent of any single customer’s preferences. High-support itemsets (like milk with vegetables) reflect combinations tied to very frequently purchased staple items, rather than a specific, deliberate pairing choice by a subset of customers — both milk and vegetables individually have very high standalone frequency (Table 1), so their pairing being common is expected rather than a surprising discovery.

Question 3: Association Rules

Applying `association_rules` (confidence ≥ 0.3) to the frequent itemsets produced 487 rules in total. Table 3 lists 10 representative rules, sorted by lift.

Table 3: Ten association rules generated from the Groceries dataset.

Antecedent	Consequent	Support	Confidence	Lift
{root vegetables, pip fruit}	{whole milk, other vegetables}	0.0055	0.353	4.716
{root vegetables, tropical fruit}	{whole milk, other vegetables}	0.0070	0.333	4.454
{citrus fruit, root vegetables}	{whole milk, other vegetables}	0.0058	0.328	4.377
{citrus fruit, whole milk, other vegetables}	{root vegetables}	0.0058	0.445	4.085
{whipped/sour cream, root vegetables}	{whole milk, other vegetables}	0.0052	0.304	4.057
{yogurt, root vegetables}	{whole milk, other vegetables}	0.0078	0.303	4.051
{herbs}	{root vegetables}	0.0070	0.431	3.956
{citrus fruit, pip fruit}	{tropical fruit}	0.0056	0.404	3.854
{whole milk, tropical fruit, other vegetables}	{root vegetables}	0.0070	0.411	3.768
{yogurt, whole milk, root vegetables}	{tropical fruit}	0.0057	0.392	3.732

Question 4: Understanding the Measures

Three rules are examined in detail:

Rule 1: **{root vegetables, pip fruit}** $\rightarrow$ **{whole milk, other vegetables}** (support 0.0055, confidence 0.353, lift 4.716)

- **Support (0.0055):** this specific four-item combination occurs in about 0.55% of all transactions — a fairly rare overall combination.
- **Confidence (0.353):** among customers who bought root vegetables and pip fruit together, about 35.3% also bought whole milk and other vegetables.
- **Lift (4.716):** customers who bought root vegetables and pip fruit were about 4.7 times more likely to also buy whole milk and other vegetables than a random customer — a strong positive association.

Rule 2: {herbs} → {root vegetables} (support 0.0070, confidence 0.431, lift 3.956)

- **Support (0.0070):** herbs and root vegetables appear together in 0.70% of transactions.
- **Confidence (0.431):** 43.1% of customers who bought herbs also bought root vegetables.
- **Lift (3.956):** buying herbs makes a customer almost 4 times more likely than average to also buy root vegetables — consistent with herbs and root vegetables both being used in similar cooking (e.g., stews, soups).

Rule 3: {other vegetables} → {whole milk} (support 0.0748, confidence 0.387, lift 1.514)

- **Support (0.0748):** this pair appears in 7.5% of all transactions — the most common two-item pairing in the dataset.
- **Confidence (0.387):** 38.7% of customers who bought other vegetables also bought whole milk.
- **Lift (1.514):** customers buying other vegetables are only about 1.5 times more likely to buy whole milk than a random customer — a much weaker association than Rules 1 and 2, despite its much higher support, since both items are so individually common that some co-occurrence is expected by chance alone.

Question 5: Find Interesting Rules

- **Highest confidence:** {tropical fruit, yogurt, root vegetables} → {whole milk} (confidence = 0.700, support = 0.0057, lift = 2.740)
- **Highest lift:** {root vegetables, pip fruit} → {whole milk, other vegetables} (lift = 4.716, support = 0.0055, confidence = 0.353)
- **Highest support:** {other vegetables} → {whole milk} (support = 0.0748, confidence = 0.387, lift = 1.514)

d. Are these the same rule? No — all three are different rules. This illustrates an important point about association rule mining: a rule can be very common (high support) without representing a particularly meaningful or surprising relationship (low lift), because it involves items that are independently very popular. Conversely, a rule can reveal a strong, meaningful relationship (high lift) while remaining relatively rare overall (low support), because it involves more specific or less commonly purchased items. No single metric alone identifies the “best” rule — support, confidence, and lift each capture a different aspect of rule quality, and a business decision typically needs to weigh all three together.

Question 6: Interpret a Rule

Selected rule: {root vegetables, pip fruit} $\rightarrow$ {whole milk, other vegetables}

In simple language: customers who purchase root vegetables and pip fruit (e.g., apples, pears) together also tend to purchase whole milk and other vegetables in the same shopping trip.

Strength of the association: this is a **strong** association. Although its support is low (0.0055, meaning this exact four-item combination is relatively rare), its lift of 4.716 indicates that customers who buy the antecedent items are almost five times more likely than an average customer to also buy the consequent items. A lift well above 1 is the key indicator of a strong, non-coincidental relationship — the low support here simply reflects that not many customers buy this particular combination of fruits and vegetables, not that the relationship itself is weak.

Question 7: Business Application

a. Products to place close together: Root vegetables and other vegetables show one of the highest supports and strong lift (support = 0.047, lift = 2.25) among simple two-item rules, suggesting the store could place these sections adjacent to each other, or ensure produce aisles group root vegetables near other vegetable displays to encourage combined purchases.

b. Possible product bundle: A “pasta night” or produce bundle of {root vegetables, pip fruit, whole milk, other vegetables} could be promoted together, given the strong lift (4.716) identified in Rule 1 — for example, a discount bundle or an in-store display combining fresh produce with dairy.

c. Recommendation system example: An online grocery ordering platform could use the rule {tropical fruit, yogurt, root vegetables} $\rightarrow$ {whole milk} (confidence 0.700) to trigger a “customers who bought these items also bought whole milk” recommendation at checkout whenever a customer’s cart contains tropical fruit, yogurt, and root vegetables together, since 70% of past customers with that combination went on to also purchase whole milk.

Question 8: Understanding Association

a. Does an association rule prove causation? No. Association rules describe **co-occurrence patterns** in the data — they show that two sets of items tend to appear together in the same transactions, but they do not establish that purchasing one set of items *causes* a customer to purchase the other. The relationship could instead be driven by a third factor (e.g., both items being needed for a common recipe, both being on promotion in the same week, or both simply being popular staple items bought by many customers for unrelated reasons).

b. Why might a rule have high confidence but low lift? This happens when the consequent item is already very popular on its own (has high overall support in the dataset), regardless of the antecedent. For example, if whole milk is purchased in 25% of all transactions, then almost any antecedent will have a reasonably high confidence of “leading to” whole milk purchases simply because whole milk is bought so frequently by everyone — not because there’s a meaningful relationship between the antecedent and whole milk specifically. Lift corrects for this baseline popularity, so a rule with high confidence but lift close to 1 indicates the apparent association is mostly just a reflection of how common the consequent already is, not a genuine pattern.

c. What does a lift value greater than 1 indicate? A lift greater than 1 indicates a **positive association** — the antecedent and consequent occur together more often than would be

expected if they were statistically independent. The further above 1, the stronger the association (lift = 1 means no association/ independence; lift < 1 would indicate a negative association, where the items are less likely to be bought together than by chance).